

NANGUAN LIN

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Education

University of Illinois Urbana-Champaign

Master of Computer Engineering

August 2024 - December 2025

Champaign, USA

East China University of Science and Technology

Bachelor of Computer Science

September 2020 - June 2024

Shanghai, China

Technical Skills

Languages: Ruby, JavaScript, TypeScript, Python, Java, C, C++, SQL, HTML/CSS

Frameworks: Ruby on Rails, Angular, React, Vue, Express, Flask, Django

Databases: PostgreSQL, MySQL, SQLite, MongoDB, Redis, AWS S3, DynamoDB

Tools: Git, Docker, AWS, GCP, Kubernetes, CI/CD, Jenkins, Ansible, Terraform

Professional Experience

One Community Global

September 2024 - January 2025

Software Developer

Remote, USA

- Developed features and resolved bugs in the Highest Good Network app using **React**, **Redux**, and **Express.js**.
- Engineered **RESTful APIs** with **Node.js** and **Express.js**, utilizing Mongoose for **MongoDB** schema validation.
- Implemented test-driven development (TDD) with Jest and integrated **CircleCI** for continuous integration workflows.

Gitea Ltd.

August 2023 - December 2023

Software Engineer Intern

Shanghai, China

- Maintained Gitea, a self-hosted all-in-one software development service written in **Go**, including **Git hosting**, **code review**, **team collaboration**, **package registry**, and **CI/CD** functionalities, earning **44k** stars on GitHub.
- Designed the built-in badge system **RESTful API**, providing SVG badges to monitor **CI/CD pipeline** status.
- Decoupled **Elasticsearch** and **Meilisearch** from default database search, significantly improving search performance.
- Improved the **unit testing** system by strengthening data consistency and integrity checks, increasing test reliability.
- Tackled complex bugs related to **database timezone** issues, **backend concurrency** problems, and **search engine rebuilding**, improving overall system stability and performance.
- Authored three comprehensive tutorials about **package versioning**, **pipeline caching** and **built-in Docker registry**, helping new developers quickly onboard and become productive.

Selected Projects

Software Project Management Tool for Agile Teams

React, Ruby on Rails, Agile, PostgreSQL

- Utilized **React** and **Ruby on Rails** to create a web project management tool tailored for agile methodology.
- Enhanced the tool with intuitive **Kanban** and **Scrum** boards, including custom workflows, sub-tasks, **EPICS**, swim lanes, and sprint planning for a practical, visual project management approach.
- Integrated **PostgreSQL** as the primary database, optimizing data storage and retrieval processes for backend.

Scalable Microservice Scaffold

Spring Boot, Kafka, Docker, Kubernetes

- Developed a modular and extensible **microservice scaffold** in **Java** using the **Spring Boot** framework, implementing essential microservices such as user management, logging, messaging, and email.
- Utilized **RESTful APIs** for efficient inter-service communication, **Kafka** for reliable asynchronous messaging, **MongoDB** as a centralized logging and storage solution, and **Eureka** for dynamic service discovery and registration.
- Containerized and orchestrated microservices using **Docker** and **Kubernetes**, ensuring high availability and scalability.

Open-source Contributions

Outline: Open-source Knowledge Base for Growing Teams

AWS S3, JavaScript

- Facilitated enhanced system performance and maintainability by spearheading the upgrade of the **S3 storage backend** to the **S3 JavaScript SDK v3**. This initiative significantly reduced latency and streamlined future code upgrades.

Dify: Open-source LLM APP Development Platform

LLM, TypeScript

- Refactored frontend LLM codebase, replacing loosely-typed literals with **TypeScript const enums** for configuration, substantially improved code readability and maintainability.

Curve: Open-source Distributed Storage System

Distributed System, Cloud-native

- Streamlined Curve's CLI using **Protobuf** and **gRPC**, introducing intuitive commands for storage management, system monitoring and data snapshots. Enhanced ease of deployment and operation across the **distributed storage system**.